IHP

SG13CMOS5L

Open Source

Process Specification

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1 General

1.1 Scope

This document describes the process specification for IHPs SG13CMOS5L technology. The backend option offers 4 thin Al metal layers and one thick Al metal layers.

1.2 List of Abbreviations

Table 1.1: List of abbreviations used within this document

Abbreviation	Explanation
IC	Integrated Circuit
IHP	Innovations for High Performance Microelectronics
NMOS	Negative Channel Metal Oxide Semiconductor
PMOS	Positive Channel Metal Oxide Semiconductor
RD	Reference Document

1.3 Reference Documents

1.4 Main Processing Sequence and Cross-Section Schematic

- Shallow trench isolation (STI)
- NWell formation
- PWell formation
- Triple Well formation
- Poly Gate formation
- Bipolar Window opening
- Collector Window opening
- Emitter opening
- Emitter Poly definition
- Base Poly definition
- nSD implant / drive
- pSD implant / drive
- Salicide formation
- Contact definition
- Metal1
- Via1
- Metal2
- Via2
- Metal3
- Via3
- Metal4
- TopVia1
- TopMetal1
- Passivation
- Parametric test

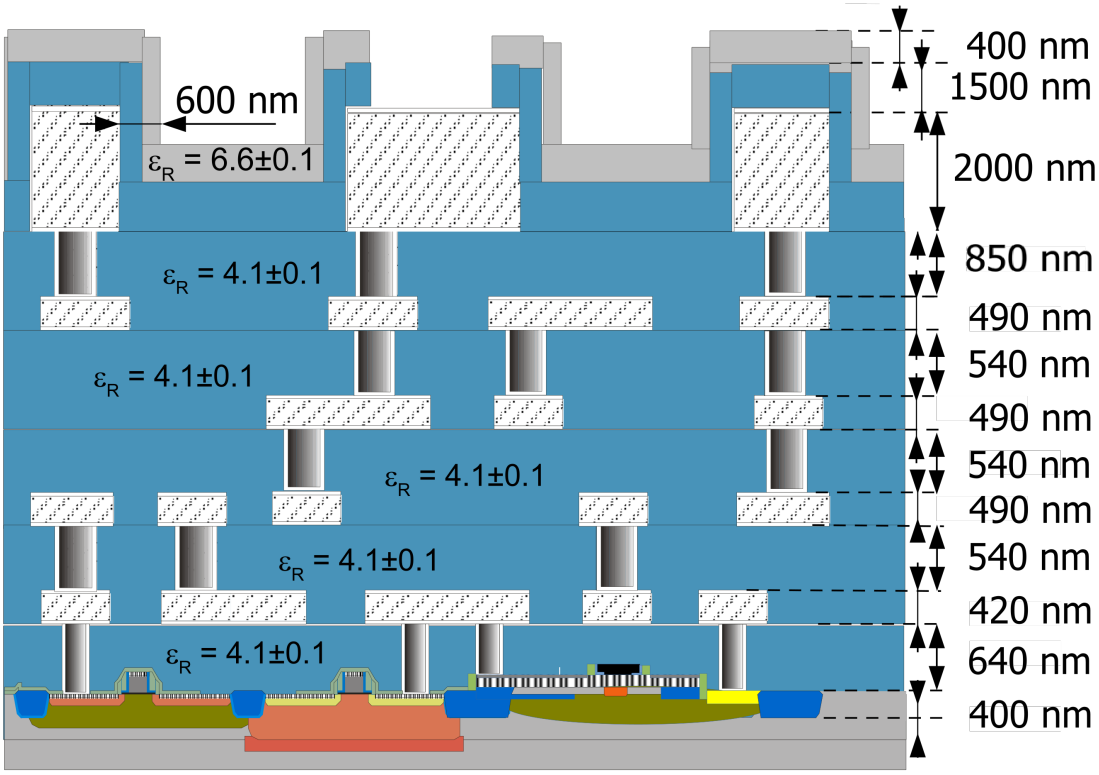

Figure 1.1: SG13CMOS5L Al process cross-section (not to scale)

Remark:
ILDO consists of oxide (590nm) and nitride (50nm).
For a homogenous ILDO with ε_R=4.1 the effective thickness corresponds to d_{eff}=620nm.

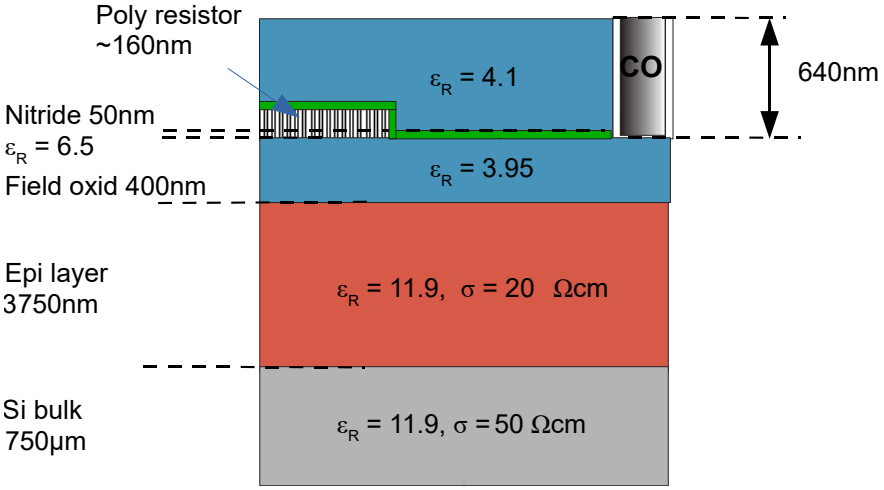

Figure 1.2: BEOL detail cross-section below Metal1 for passive modeling (not to scale)

1.5 Process Control

Several geometrical and electrical parameters are measured for process control purposes. Electrical measurements are typically performed at T_0 = $27\,°$C ($300\,$K). Coefficients describing the temperature behavior of parameters are extracted for the temperature range $-40\,°\text{C} < \text{T} < 125\,°\text{C}$.

Process control parameters are assigned to one of the following categories:

1.5.1 Pass/Fail Parameters

Pass/fail parameters are used for wafer selection during the wafer fabrication process. Pass/Fail parameters are measured on each wafer and are characterized by SPEC limits in **bold red**.

1.5.2 Information Parameters

Information parameters are provided in order to increase the knowledge about the process. These parameters do not lead to wafer scrap. There are two groups of information parameters:

The first group of information parameters is measured, identically to the pass/fail parameters, on each wafer. These parameters are characterized by SPEC limits in **bold blue**.

The second group of information parameters is not measured on each wafer. This group includes, for example, the layer thickness values measured during the wafer fabrication process or taken from REM cross-sections, and the coefficients describing the temperature, voltage and matching behavior of active and passive devices. This group of information parameters is given in *italic blue*.

Notes:

- The process control transistor parameters must not be used for circuit simulation purposes. They are often extracted from simplified model equations in order to increase the speed of the measurements. Special circuit simulation transistor parameters are provided in the models. Those are extracted from the complete set of model equations in order to give the best fit of the entire characteristic for all operating points. Therefore, process control transistor parameters may differ from their corresponding circuit simulation parameters.
- It is strongly recommended that a design shall rely only on pass/fail parameters.

1.6 Wafer Reject Criteria

Pass/fail parameters are measured on nine (or more) sites, distributed uniformly across the wafer, see Fig. 1.3. At least 2/3 of the measured sites (min. 6 sites) must pass all pass/fail criteria in order to consider the tested wafer as pass.

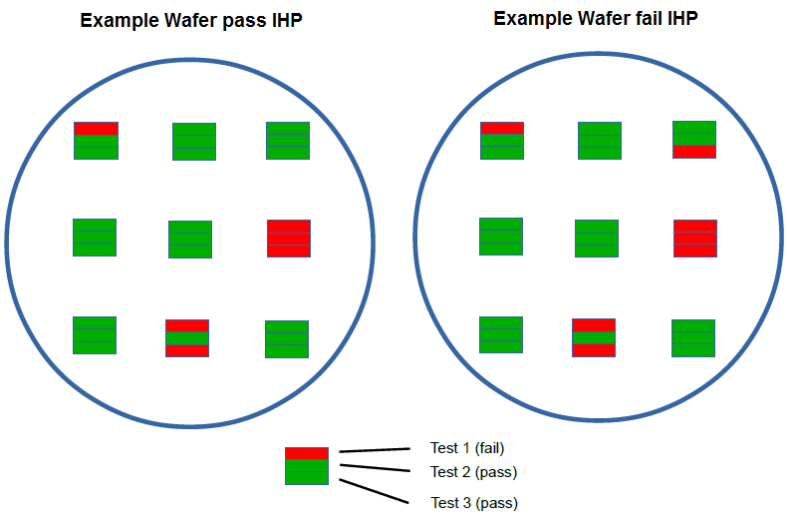

Figure 1.3: IHP PCM structures placements

2 Process Control Parameters

2.1 MOS Parameters

2.1.1 NMOS

$V_{GS} \leq 1.65$ V @ 125°C[1]

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Threshold Voltage Short Channel Device	VTN10x013	V	**0.43**	**0.50**	**0.55**	[A.a1]	WxL = 10 x 0.13 μm^2
Threshold Voltage Long Channel Device	VTN10x10	V	**0.16**	**0.20**	**0.24**	[A.a1]	WxL = 10 x 10 μm^2
Threshold Voltage Small Channel Device	VTN015x013	V	**0.4**	**0.54**	**0.68**	[A.a1]	WxL = 0.15 x 0.13 μm^2
Saturation Current Short Channel Device	IDSN013	$\mu A/\mu m$	**380**	**480**	**600**	[A.b1]	WxL = 10 x 0.13 μm^2
Off-Current Short Channel Device	IOFFN013	LOG10 ($A/\mu m$)		**-10**	**-9**	[A.c1]	WxL = 10 x 0.13 μm^2
Drain Induced Barrier Lowering 0.1/1.2V	DIBLN013	mV/V	**20**	**50**	**80**	[A.d1]	WxL = 10 x 0.13 μm^2
Sub Threshold Slope	SSN013	mV/dec	**76**	**82**	**88**	[A.e]	WxL = 10 x 0.13 μm^2
Breakdown Voltage	BVDSSN013	V	**2.0**	**2.7**		[A.f1]	WxL = 10 x 0.13 μm^2
Effective Channel Length	LEFFN013	μm	**0.10**	**0.14**	**0.19**	[A.g1]	WxL = 10 x 0.13 μm^2
Effective Channel Width	WEFFN015	μm	**0.09**	**0.15**	**0.22**	[A.h1]	WxL = 0.15 x 0.13 μm^2
Miller Capacitance NMOS	CMILLERN	$fF/\mu m$	**0.32**	**0.36**	**0.40**	[A.k1]	
Junction Capacitance NMOS	CJUNCTIONN	$fF/\mu m^2$	**0.9**	**0.95**	**1.0**	[A.k]	
Junction Breakdown	BVNPW	V		*12*		[A.f3]	

[1] 1 FIT in 10 years for TDDB: for details see SG13G2 Qualification Report

2.1.2 PMOS

$V_{GS} \leq 1.65$ V @ 125°C[2]

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Threshold Voltage Short Channel Device	VTP10x013	V	**-0.53**	**-0.47**	**-0.41**	[A.a1]	WxL = 10 x 0.13 μm^2
Threshold Voltage Long Channel Device	VTP10x10	V	**-0.41**	**-0.36**	**-0.31**	[A.a1]	WxL = 10 x 10 μm^2
Threshold Voltage Small Channel Device	VTP015x013	V	**-0.58**	**-0.48**	**-0.38**	[A.a1]	WxL = 0.15 x 0.13 μm^2
Saturation Current Short Channel Device	IDSP013	$\mu A/\mu m$	**-270**	**-215**	**-170**	[A.b1]	WxL = 10 x 0.13 μm^2
Off-Current Short Channel Device	IOFFP013	LOG10 ($A/\mu m$)		**-10.3**	**-9.3**	[A.c1]	WxL = 10 x 0.13 μm^2
Drain Induced Barrier Lowering 0.1/1.2V	DIBLP013	mV/V	**25**	**50**	**75**	[A.d1]	WxL = 10 x 0.13 μm^2
Sub Threshold Slope	SSP013	mV/dec	**-87**	**-81**	**-75**	[A.e]	WxL = 10 x 0.13 μm^2
Breakdown Voltage	BVDSSP013	V		**-2.9**	**-2.2**	[A.f1]	WxL = 10 x 0.13 μm^2
Effective Channel Length	LEFFP013	μm	**0.07**	**0.10**	**0.13**	[A.g1]	WxL = 10 x 0.13 μm^2
Effective Channel Width	WEFFP015	μm	**0.17**	**0.24**	**0.31**	[A.h1]	WxL = 0.15 x 0.13 μm^2
Miller Capacitance NMOS	CMILLERP	$fF/\mu m$	**0.31**	**0.35**	**0.39**	[A.k1]	
Junction Capacitance NMOS	CJUNCTIONP	$fF/\mu m^2$	**0.8**	**0.85**	**0.9**	[A.k]	
Junction Breakdown	BVNPW	V		*-12*		[A.f3]	

[2] 1 FIT in 10 years for TDDB: for details see SG13G2 Qualification Report

2.1.3 iNMOS

$V_{GS} \leq 1.65$ V @ 125°C[3]

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Threshold Voltage Short Channel Device	VTNI10x013	V	**0.43**	**0.50**	**0.55**	[A.a1]	WxL = 10 x 0.13 μm^2
Saturation Current Short Channel Device	IDSNI013	μA/μm	**380**	**480**	**600**	[A.b1]	WxL = 10 x 0.13 μm^2
Off-Current Short Channel Device	IOFFNI013	LOG10 (A/μm)		**-10**	**-9**	[A.c1]	WxL = 10 x 0.13 μm^2
Drain Induced Barrier Lowering 0.1/1.2V	DIBLNI013	mV/V	**20**	**50**	**80**	[A.d1]	WxL = 10 x 0.13 μm^2
Sub Threshold Slope	SSNI013	mV/dec	**76**	**82**	**88**	[A.e]	WxL = 10 x 0.13 μm^2
Breakdown Voltage	BVDSSNI013	V	**2.0**	**2.7**		[A.f1]	WxL = 10 x 0.13 μm^2

[3] 1 FIT in 10 years for TDDB: for details see SG13G2 Qualification Report

2.1.4 HV-NMOS

$V_{GS} \leq 3.3$ V (Maximum) @ 27°C for $L_G \geq 0.6\ \mu m$[4]

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Threshold Voltage Short Channel Device	VTNHV10x045	V	**0.63**	**0.70**	**0.77**	[A.a2]	WxL = 10 x 0.45 μm^2
Threshold Voltage Long Channel Device	VTNHV10x10	V	**0.65**	**0.69**	**0.73**	[A.a2]	WxL = 10 x 10 μm^2
Threshold Voltage Small Channel Device	VTNHV030x045	V	**0.59**	**0.67**	**0.75**	[A.a2]	WxL = 0.30 x 0.45 μm^2
Saturation Current Short Channel Device	IDSNHV045	$\mu A/\mu m$	**480**	**560**	**640**	[A.b2]	WxL = 10 x 0.45 μm^2
Off-Current Short Channel Device	IOFFNHV045	LOG10 ($A/\mu m$)		**-12.5**	**-11**	[A.c2]	WxL = 10 x 0.45 μm^2
Drain Induced Barrier Lowering 0.1/1.2V	DIBLNHV045	mV/V	**0**	**15**	**30**	[A.d2]	WxL = 10 x 0.45 μm^2
Sub Threshold Slope	SSNHV013	mV/dec	**72**	**84**	**96**	[A.e]	WxL = 10 x 0.45 μm^2
Breakdown Voltage	BVDSSNHV013	V	**5.3**	**6.1**		[A.f2]	WxL = 10 x 0.45 μm^2
Effective Channel Length	LEFFNHV045	μm	**0.26**	**0.31**	**0.36**	[A.g2]	WxL = 10 x 0.45 μm^2
Effective Channel Width	WEFFNHV030	μm	**0.23**	**0.28**	**0.33**	[A.h2]	WxL = 0.30 x 0.45 μm^2
Miller Capacitance NMOS	CMILLERNHV	$fF/\mu m$	**0.42**	**0.45**	**0.48**	[A.k1]	
Junction Capacitance NMOS	CJUNCTIONNHV	$fF/\mu m^2$	**0.74**	**0.80**	**0.86**	[A.k]	
Junction Breakdown	BVNPWHV	V		*12*		[A.f3]	

[4] 1 FIT in 10 years for HCI: for details see SG13G2 Qualification Report

2.1.5 HV-PMOS

$V_{GS} \leq 3.3$ V (Maximum) @ 27°C for $L_G \geq 0.5\ \mu m$[5]

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Threshold Voltage Short Channel Device	VTPHV10x04	V	**-0.71**	**-0.65**	**-0.59**	[A.a2]	WxL = 10 x 0.4 μm^2
Threshold Voltage Long Channel Device	VTPHV10x10	V	**-0.78**	**-0.70**	**-0.64**	[A.a2]	WxL = 10 x 10 μm^2
Threshold Voltage Small Channel Device	VTPHV03x04	V	**-0.71**	**-0.64**	**-0.57**	[A.a2]	WxL = 0.3 x 0.4 μm^2
Saturation Current Short Channel Device	IDSPHV04	$\mu A/\mu m$	**-290**	**-240**	**-190**	[A.b2]	WxL = 10 x 0.4 μm^2
Off-Current Short Channel Device	IOFFPHV04	LOG10 ($A/\mu m$)		**-12.5**	**-11.5**	[A.c2]	WxL = 10 x 0.4 μm^2
Drain Induced Barrier Lowering 0.1/1.2V	DIBLPHV04	mV/V	**0**	**5**	**15**	[A.d2]	WxL = 10 x 0.4 μm^2
Sub Threshold Slope	SSPHV04	mV/dec	**-102**	**-92**	**-82**	[A.e]	WxL = 10 x 0.4 μm^2
Breakdown Voltage	BVDSSPHV04	V		**-6.3**	**-5.3**	[A.f2]	WxL = 10 x 0.4 μm^2
Effective Channel Length	LEFFPHV04	μm	**0.24**	**0.30**	**0.36**	[A.g2]	WxL = 10 x 0.4 μm^2
Effective Channel Width	WEFFPHV03	μm	**0.26**	**0.33**	**0.40**	[A.h2]	WxL = 0.3 x 0.4 μm^2
Miller Capacitance NMOS	CMILLERPHV	$fF/\mu m$	**0.32**	**0.35**	**0.38**	[A.k1]	
Junction Capacitance NMOS	CJUNCTIONPHV	$fF/\mu m^2$	**0.74**	**0.80**	**0.86**	[A.k]	
Junction Breakdown	BVNPWhv	V		*-12*		[A.f3]	

[5] 1 FIT in 10 years for HCI: for details see SG13G2 Qualification Report

2.1.6 HV-iNMOS

$V_{GS} \leq 3.3$ V (Maximum) @ 27°C for $L_G \geq 0.6\ \mu m$[6]

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Threshold Voltage Short Channel Device	VTNIHV10x045	V	**0.63**	**0.70**	**0.77**	[A.a2]	WxL = 10 x 0.45 μm^2
Saturation Current Short Channel Device	IDSNIHV045	$\mu A/\mu m$	**480**	**560**	**640**	[A.b2]	WxL = 10 x 0.45 μm^2
Off-Current Short Channel Device	IOFFNIHV045	LOG10 ($A/\mu m$)		**-12.5**	**-11**	[A.c2]	WxL = 10 x 0.45 μm^2
Drain Induced Barrier Lowering 0.1/1.2V	DIBLNIHV045	mV/V	**0**	**15**	**30**	[A.d2]	WxL = 10 x 0.45 μm^2
Sub Threshold Slope	SSNIHV045	mV/dec	**72**	**84**	**96**	[A.e]	WxL = 10 x 0.45 μm^2
Breakdown Voltage	BVDSSNIHV045	V	**5.2**	**6.1**		[A.f2]	WxL = 10 x 0.45 μm^2

[6] 1 FIT in 10 years for HCI: for details see SG13G2 Qualification Report

2.2 Passive Device Parameters

2.2.1 Rsil

Rsil utilizes **salicided, n-doped gate polysilicon** as resistor material.

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Sheet Resistance	RSNRSIL	Ω	**6.2**	**7.0**	**7.8**	[A.i]	
Line Width Delta	DWRSIL	nm	**-20**	**10**	**40**	[A.i]	
Temperature Coefficients	TC1NRSIL TC2NRSIL	ppm/K ppm/K^2		*3100 0.3*		[A.af]	
Matching Coefficients	MATRSIL1 MATRSIL2	nm		*6 1.4*		[A.ac]	
Metal-to-Body-Resistance	RCRSIL	Ω* µm		*4.5*		[A.ae]	

Maximum Current Density for Rsil is limited by contacts, please refer section 3.5.

2.2.2 Rppd

Rppd utilizes **unsalicided, p-doped gate polysilicon** as resistor material. For realizing precision resistors, a line width of 2µm or higher is recommended.

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Sheet Resistance	RSRSRPPD	Ω	**235**	**260**	**285**	[A.i]	
Line Width Delta	DWRPPD	nm	**-24**	**6**	**36**	[A.i]	
Temperature Coefficients	TC1NRPPD TC2NRPPD	ppm/K ppm/K^2		*170 0.4*		[A.af]	
Matching Coefficients	MATRPPD	nm		*15*		[A.ac]	
Metal-to-Body-Resistance	RCRPPD	Ω* µm		*35*		[A.ae]	
Temperature Coefficient Metal-to-Body-Resistance	TC3NRPPD	ppm/K		*-950*			
Max. Current Density	IMRPPD	mA/µm			*1.2*		11 years @ 105°C

2.2.3 Rhigh

Rhigh utilizes **unsalicided, partially compensated gate polysilicon** as resistor material.

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Sheet Resistance	RSRHIGH	Ω	**1160**	**1360**	**1560**	[A.i]	
Line Width Delta	DWRHIGH	nm	**-80**	**-40**	**0**	[A.i]	
Temperature Coefficients	TC1NRHIGH TC2NRHIGH	ppm/K ppm/K^2		*-2300*	*2.1*	[A.af]	
Matching Coefficients	MATRHIGH	nm		*48*		[A.ac]	
Metal-to-Body-Resistance	RCRHIGH	Ω* µm		*80*		[A.ae]	
Max. Current Density	IMRHIGH	mA/µm			*0.6*		11 years @ 105°C

2.2.4 S-Varicap

Thick Gate Oxide.

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Capacitance -3	SVAR_m3	fF/μm^2	**20.5**	**23**	**25.5**		10x (3.74x0.3)μm^2 @15.8GHz
Capacitance 0	SVAR_0	fF/μm^2	**32**	**35.3**	**37.5**		10x (3.74x0.3)μm^2 @15.8GHz
Capacitance +3	SVAR_3	fF/μm^2	**37.5**	**39.5**	**41.5**		10x (3.74x0.3)μm^2 @15.8GHz
Q Factor -3	QFACTOR_m3		**50**	**62**	**75**		10x (3.74x0.3)μm^2 @15.8GHz
Q Factor 0	QFACTOR_0		**35**	**43**	**50**		10x (3.74x0.3)μm^2 @15.8GHz
Q Factor 3	QFACTOR_3		**35**	**43**	**50**		10x (3.74x0.3)μm^2 @15.8GHz

3 Al BEOL

3.1 Resistances

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Substrate Resistivity	RSBLK	Ωcm	*37.5*	*50*	*62.5*		WAFPR3763
Salicided GatPoly (n^+)							refer section 2.2.1
Unsalicided GatPoly (p^+)							refer section 2.2.2
Unsalicized GatPoly (n^+)							refer section 2.2.3
Unsalicided nSD-Activ Sheet Resistance	RSNSD0	Ω	**55**	**67**	**79**		
Unsalicided pSD-Activ Sheet Resistance	RSPSD0	Ω	**69**	**79**	**89**		
Metal1 Snake Sheet Resistance	SNAKEM1	mΩ	**90**	**115**	**145**		width = 0.16 µm
Metal2 Snake Sheet Resistance	SNAKEM2	mΩ	**70**	**88**	**110**		width = 0.20 µm
Metal3 Snake Sheet Resistance	SNAKEM3	mΩ	**70**	**88**	**110**		width = 0.20 µm
Metal4 Snake Sheet Resistance	SNAKEM4	mΩ	**70**	**88**	**110**		width = 0.20 µm
TopMetal1 Snake Sheet Resistance	SNAKETM1	mΩ	**14**	**18**	**22**		width = 1.5 µm
Metal1 Sheet Resistance	RSMET1	mΩ	**85**	**110**	**135**	[A.i]	
Metal2 Sheet Resistance	RSMET2	mΩ	**73**	**88**	**103**	[A.i]	
Metal3 Sheet Resistance	RSMET3	mΩ	**73**	**88**	**103**	[A.i]	
Metal4 Sheet Resistance	RSMET4	mΩ	**73**	**88**	**103**	[A.i]	
TopMetal1 Sheet Resistance	RSTM1	mΩ	**15**	**18**	**21**	[A.i]	

3.2 Line Width Deltas

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Metal1 Line Width Delta	DWMET1	nm	**-64**	**-24**	**16**	[A.i]	
Metal2 Line Width Delta	DWMET2	nm	**-56**	**-16**	**24**	[A.i]	
Metal3 Line Width Delta	DWMET3	nm	**-56**	**-16**	**24**	[A.i]	
Metal4 Line Width Delta	DWMET4	nm	**-56**	**-16**	**24**	[A.i]	
TopMetal1 Line Width Delta	DWTM1	nm	**-300**	**-100**	**100**	[A.i]	

3.3 Temperature Coefficients

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Metal1 Sheet Resistance Temperature Coefficient	TC1RSMET1	ppm/K		*3400*		[A.af]	
Metal2 Sheet Resistance Temperature Coefficient	TC1RSMET2	ppm/K		*3500*		[A.af]	
Metal3 Sheet Resistance Temperature Coefficient	TC1RSMET3	ppm/K		*3500*		[A.af]	
Metal4 Sheet Resistance Temperature Coefficient	TC1RSMET4	ppm/K		*3500*		[A.af]	
TopMetal1 Sheet Resistance Temperature Coefficient	TC1RSTM1	ppm/K		*3700*		[A.af]	

3.4 Contact and Via Resistances

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Metal1 to Silicide on nSD-Activ	RCM1NSD	Ω/CNT	**8**	**17**	**22**		V = 1V, 93740 contact chain
Metal1 to Silicide on pSD-Activ	RCM1NPD	Ω/CNT	**8**	**17**	**22**		V = 1V, 93740 contact chain
Metal1 to Silicide on GatPoly (n^+)	RCM1NPLY	Ω/CNT	**8**	**15**	**20**		V = 1V, 98566 contact chain
Metal1 to Silicide on GatPoly (p^+)	RCM1PPLY	Ω/CNT	**8**	**15**	**20**		V = 1V, 98566 contact chain
Metal2 - Metal1	RVIA1	Ω/VIA	**5**	**9**	**20**		V = 1V, 103840 contact chain
Metal3 - Metal2	RVIA2	Ω/VIA	**5**	**9**	**20**		V = 1V, 103840 contact chain
Metal4 - Metal3	RVIA3	Ω/VIA	**5**	**9**	**20**		V = 1V, 103840 contact chain
TopMetal1 - Metal4	tbd	Ω/VIA	**tbd**	**tbd**	**tbd**		tbd

3.5 Maximum Current Densities

(11 years @ 105°C)

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Contact	JMAXCNT	mA/Cnt			*0.3*	[A.v]	
Metal1	IMAXM1	mA			*0.36*	[A.v]	w = 0.16 ... 0.36 μm
Metal1	JMAXM1	mA/μm			*1*	[A.v]	w > 0.36 μm
Via1	JMAXVIA1	mA/Via			*0.4*	[A.v]	
Metal2	IMAXM2	mA			*0.6*	[A.v]	w = 0.2 ... 0.3 μm
Metal2	JMAXM2	mA/μm			*2*	[A.v]	w > 0.3 μm
Via2	JMAXVIA2	mA/Via			*0.4*	[A.v]	
Metal3	IMAXM3	mA			*0.6*	[A.v]	w = 0.2 ... 0.3 μm
Metal3	JMAXM3	mA/μm			*2*	[A.v]	w > 0.3 μm
Via3	JMAXVIA3	mA/Via			*0.4*	[A.v]	
Metal4	IMAXM4	mA			*0.6*	[A.v]	w = 0.2 ... 0.3 μm
Metal4	JMAXM4	mA/μm			*2*	[A.v]	w > 0.3 μm
TopVia1	JMAXTVIA1	mA/Via			*1.4*	[A.v]	
TopMetal1	JMAXM6	mA/μm			*15*	[A.v]	

3.6 Layer Thickness Values

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Shallow Trench Isolator Thickness	TSTI	nm		*400*		[A.ag]	
Gate Polysilicon Thickness	TGATPOLY	nm	*150*	*160*	*170*	[A.w]	
Thickness of Gate Oxide	TGOXNW	nm	**2.25**	**2.45**	**2.65**	[A.x]	
Thickness of Gate Oxide HV-MOS	TGOX1NW	nm	**6.8**	**7.3**	**7.8**	[A.x]	
Thickness of Gate Oxide	TGOXPW	nm	**2.45**	**2.65**	**2.85**	[A.x]	
Thickness of Gate Oxide HV-MOS	TGOX1PW	nm	**7.0**	**7.5**	**8.0**	[A.x]	
Isolator Thickness between Activ and Metal1	TILD0	nm		*640*		[A.ag]	[A.w]
Metal1 Layer Thickness	TMET1	nm		*420*		[A.ag]	
Isolator Thickness between Metal1 and Metal2	TILD1	nm		*540*		[A.ag]	[A.w]
Metal2 Layer Thickness	TMET2	nm		*490*		[A.ag]	
Isolator Thickness between Metal2 and Metal3	TILD2	nm		*540*		[A.ag]	[A.w]
Metal3 Layer Thickness	TMET3	nm		*490*		[A.ag]	
Isolator Thickness between Metal3 and Metal4	TILD3	nm		*540*		[A.ag]	[A.w]
Metal4 Layer Thickness	TMET4	nm		*490*		[A.ag]	
Isolator Thickness between Metal4 and TopMetal1	TILDTM1	nm		*850*		[A.ag]	[A.w]
TopMetal1 Layer Thickness	TTM1	nm		*2000*		[A.ag]	
Thickness Values of Passivation Layers	TPASS1 TPASS2	nm		*1500* *400*		[A.ag]	Oxide layer SiN layer

3.7 Parasitic Capacitances

Parameter	Name	Unit	Min	Tar-get	Max	Meas. Cond.	Comment
Metal1 - Activ Area Capacitance	CAMET1ACT	aF/μm^2	**49**	**59**	**69**	[A.aq]	A = 250 x 1200 μm^2
Metal1 - Substrate Area Capacitance	CAMET1SUB	aF/μm^2	**31**	**37**	**43**	[A.aq]	A = 250 x 1200 μm^2
Metal1 - Metal2 Area Capacitance	CAMET1/2	aF/μm^2	**54**	**68**	**82**	[A.aq]	A = 250 x 1200 μm^2
Metal2 - Metal3 Area Capacitance	CAMET2/3	aF/μm^2	**54**	**68**	**82**	[A.aq]	A = 250 x 1200 μm^2
Metal3 - Metal4 Area Capacitance	CAMET3/4	aF/μm^2	**54**	**68**	**82**	[A.aq]	A = 250 x 1200 μm^2
Metal4 - TopMetal1 Area Capacitance	tbd	aF/μm^2	**tbd**	**tbd**	**tbd**	[A.aq]	A = 250 x 1200 μm^2

Appendix A: Measurement Conditions

[A.aq] **capacitance of metal layer to active or substrate**
v = 0 v; f = 100 khz.

[A.ac] **resistor matching coefficient k**
measuring the matching behavior of resistor pairs, differing in the resistor area, are investigated.
the least square fit δ(dr/r) vs. inverse square root of area is estimated: $\delta(dr/r) = k \times a^{-0.5}$.

[A.af] **resistor temperature coefficients**
measurement of r as a function of temperature in the range −40 to 125.
temperature coefficients t_{C1} and t_{C2}; t_0 = 300k
$r(t) = r(t_0) * [1 + t_{C1} * (t - t_0) + t_{C2} * (t - t_0)^2]$.

[A.ae] **salicide to gatpoly contact resistance**
r_{RES} = (2* r_{CONT} / # of contacts) + (2 x r_{C2POLY} / resistor width) + r_{POLY}
note that this formula is only valid for resistors with same width of unsalicided and salicided regions
r_{RES}: Total resistance of a resistor; r_{CONT}: Contribution of a single w-plug (typically < 1ω)
r_{C2POLY}: Salicide to gatpoly contact resistance
r_{POLY}: Contribution of the unsalicided polysilicon stripe (= r_S x l / w).

[A.ag] **sem cross section analysis**.

[A.a1] **threshold voltage:**
v_{DS} = +0.05 v / -0.05 v (n-channel / p-channel device); v_{BS} = 0 v;
v_T is extrapolated from the maximum slope of the active transfer characteristic region.
a linear regression is performed to find this slope. $v_T = v_0 - v_{DS}/2$;
v_0 is the gate voltage intercept of the slope.
isolated nmos: V_{nWell} = 0 v.

[A.a2] **threshold voltage:**
v_{DS} = +0.1 v / -0.1 v (n-channel / p-channel device); v_{BS} = 0v;
v_T is extrapolated from the maximum slope of the active transfer characteristic region.
a linear regression is performed to find this slope. $v_T = v_0 - v_{DS}/2$;
v_0 is the gate voltage intercept of the slope.
isolated nmos: V_{nWell} = 0 v.

[A.b1] **saturation current:**
$v_{DS} = v_{GS}$ = +1.2 v / -1.2 v (n-channel / p-channel device); v_{BS} = 0 v
isolated nmos: V_{nWell} = 0 v.

[A.b2] **saturation current:**
$v_{DS} = v_{GS}$ = +3.3 v / -3.3 v (n-channel / p-channel device); v_{BS} = 0 v
isolated nmos: V_{nWell} = 0 v.

[A.c1] **off-current:**
v_{DS} = +1.2 v / -1.2 v (n-channel / p-channel device); $v_{GS} = v_{BS}$ = 0 v; i_D measured
isolated nmos: V_{nWell} = 0 v.

[A.c2] **off-current:**
v_{DS} = +3.3 v / -3.3 v (n-channel / p-channel device); $v_{GS} = v_{BS}$ = 0 v; i_D measured
isolated nmos: V_{nWell} = 0 v.

[A.d1] **drain induced barrier lowering:**
dibl = $[v_{GS}(v_{DS1}) - v_{GS}(v_{DS2})]/(v_{DS1}-v_{DS2})$ at i_D = jss*w/l
v_{DS1} = +0.1 v; v_{DS2} = +1.2 v; v_{BS} = 0 v; jss = 0.5 na (n-channel device)
v_{DS1} = -0.1 v; v_{DS2} = -1.2 v; v_{BS} = 0 v; jss = 0.1 na (p-channel device).

[A.d2] **drain induced barrier lowering:**
dibl = [$v_{GS}(v_{DS1}) - v_{GS}(v_{DS2})$]/($v_{DS1}$-$v_{DS2}$) at i_D = jss*w/l
v_{DS1} = +0.1 v; v_{DS2} = +3.3 v; v_{BS} = 0 v; jss = 1 na (n-channel device)
v_{DS1} = -0.1 v; v_{DS2} = -3.3 v; v_{BS} = 0 v; jss = 0.4 na (p-channel device).

[A.e] **sub threshold slope:**
v_{DS} = +0.1 v / -0.1 v (n-channel / p-channel device); v_{BS} = 0 v
the slope is estimated from the two drain currents i_{D1} = jss1*w/l and i_{D2} = jss2*w/l
jss1 = 0.5 na, iss2 = 5 na (n-channel device);
jss2 = 0.1 na, iss2 = 1 na (p-channel device);.

[A.f3] **breakdown voltage:**
j = 1 ma / cm^2 for s/d to body diode surrounded by sti.

[A.f1] **breakdown voltage:**
v_{GS} = v_{BS} = 0 v; i_D = i_{D0} = +10 na/µm / -10 na/µm (n-channel / p-channel device).

[A.f2] **breakdown voltage:**
v_{GS} = v_{BS} = 0 v; i_D = i_{D0} = +100 pa/µm / -100 pa/µm (n-channel / p-channel device).

[A.g1] **effective channel length:**
extrapolated from linear regions of transistors with gate lengths of 0.13 µm, 0.18 µm;
v_{DS} = +0.05 v / -0.05 v (n-channel / p-channel device)
method (p. i. suciu and r. l. johnston, ”experimental derivation of the source and drain resistance of mos transistors,” in ieee transactions on electron devices, vol. 27, no. 9, pp. 1846-1848, 1980)
basic equations:
(1) $i_D = \beta(v_{GSeff} - v_{DS}/2)\, v_{DS}$
(2) $v_{GSeff} = v_{GS} - v_T$
(3) $\mu = \mu_0 / (1 + u_0\, v_{GSeff})$
(4) $1/\beta(v_{GSeff}) = (l - \delta l) / (w\, \mu c_{OX}) = (l - \delta l)\,(1 + u_0\, v_{GSeff}) / (w\, \mu_0\, c_{OX})$
(5) $1/\beta_0(l) = (l - \delta l) / (w\, \mu_0\, c_{OX})$
first step: according to equ.(4), linear extrapolation of $1/\beta_0 = 1/\beta(v_{GSeff} = 0)$ from two working points v_{GSeff1} = +1v / -1v; v_{GSeff2} = +1.5v / -1.5v (n-channel / p-channel device) for all transistors with different gate length
second step: linear regression of δl from all $1/\beta_0(l)$ numbers according to equ.(5).

[A.g2] **effective channel length:**
extrapolated from linear regions of transistors with gate lengths of 0.33 µm (only sg13s); 0.45 μm, 10 μm (only sg13s), 0.4 μm, 10 μm;
v_{DS} = +0.1 v / -0.1 v (n-channel / p-channel device)
method (p. i. suciu and r. l. johnston, ”experimental derivation of the source and drain resistance of mos transistors,” in ieee transactions on electron devices, vol. 27, no. 9, pp. 1846-1848, 1980)
basic equations:
(1) $i_D = \beta(v_{GSeff} - v_{DS}/2)\, v_{DS}$
(2) $v_{GSeff} = v_{GS} - v_T$
(3) $\mu = \mu_0 / (1 + u_0\, v_{GSeff})$
(4) $1/\beta(v_{GSeff}) = (l - \delta l) / (w\, \mu c_{OX}) = (l - \delta l)\,(1 + u_0\, v_{GSeff}) / (w\, \mu_0\, c_{OX})$
(5) $1/\beta_0(l) = (l - \delta l) / (w\, \mu_0\, c_{OX})$
first step: according to equ.(4), linear extrapolation of $1/\beta_0 = 1/\beta(v_{GSeff} = 0)$ from two working points v_{GSeff1} = +1v / -1v; v_{GSeff2} = +1.5v / -1.5v (n-channel / p-channel device) for all transistors with different gate length
second step: linear regression of δl from all $1/\beta_0(l)$ numbers according to equ.(5).

[A.h1] **effective channel length:**
extrapolated from linear regions of transistors with gate lengths of 0.15 μm, 10.0 μm; (not isolated nmos); v_{DS} = +0.05 v / -0.05 v (n-channel / p-channel device)
basic equations:
(1) $i_D = \beta(v_{GSeff} - v_{DS}/2)\, v_{DS}$
(2) $v_{GSeff} = v_{GS} - v_T$

(3) $\mu = \mu_0 / (1 + u_0\ v_{GSeff})$
(4) $1/\beta(v_{GSeff}) = l / [(w - \delta w)\ \mu c_{OX}] = [l\ (1 + u_0\ v_{GSeff})] / [(w - \delta w)\ \mu_0\ c_{OX}]$
(5) $\beta_0(w) = (w - \delta w)\ \mu_0\ c_{OX} / l$
first step: according to equ.(4), linear extrapolation of $1/\beta_0 = 1/\beta(v_{GSeff} = 0)$ from two working points v_{GSeff1} = +1v / -1v; v_{GSeff2} = +1.5v / -1.5v (n-channel / p-channel device) for all transistors with different gate width
second step: linear regression of δw from all $\beta_0(w)$ numbers according to equ.(5).

[A.h2] **effective channel length:**
extrapolated from linear regions of transistors with gate lengths of 0.3 μm, 10.0 μm;
v_{DS} = +0.1 v / -0.1 v (n-channel / p-channel device)
basic equations:
(1) $i_D = \beta(v_{GSeff} - v_{DS}/2)\ v_{DS}$
(2) $v_{GSeff} = v_{GS} - v_T$
(3) $\mu = \mu_0 / (1 + u_0\ v_{GSeff})$
(4) $1/\beta(v_{GSeff}) = l / [(w - \delta w)\ \mu c_{OX}] = [l\ (1 + u_0\ v_{GSeff})] / [(w - \delta w)\ \mu_0\ c_{OX}]$
(5) $\beta_0(w) = (w - \delta w)\ \mu_0\ c_{OX} / l$
first step: according to equ.(4), linear extrapolation of $1/\beta_0 = 1/\beta(v_{GSeff} = 0)$ from two working points v_{GSeff1} = +1v / -1v; v_{GSeff2} = +1.5v / -1.5v (n-channel / p-channel device) for all transistors with different gate width
second step: linear regression of δw from all $\beta_0(w)$ numbers according to equ.(5).

[A.i] **sheet resistance, line width delta**
the sheet resistance r_S and the line width delta (δw) values are calculated from the resistances of two resistors r_1 and r_N:
- r_1: one single stripe with the dimension w x l
- r_N: N stripes in parallel with dimensions (w/n) x l each
formulas: $R_1 = r_S\ l / (w + \delta w)$ and $r_N = r_S\ (l/n) / (w/n + \delta w)$
voltage across both resistors in series: 0.5v.

[A.k] **capacitance measurements, area capacitance**
v = 0 v; f = 100 khz; area capacitance = $c_{measure}$ / area.

[A.k1] **capacitance measurements mim, area capacitance**
v = 2 v applied to top plate; f = 100 khz.

[A.v] **maximum current density of metal lines, contacts and vias**
the maximum current density is determined via electromigration measurements as part of the process qualification procedure. the current density values given were estimated to reach less than 0.01 % failure in 11 years of operation at 105°c (for sg13rh: 20 years at 125°c).

[A.w] **optical layer thickness**
the thickness of the layer is measured by means of optical interferometry / ellipsometry.

[A.x] **gateoxide thickness measurement using cox extrapolation**
measured capacitance corrected for offset and miller capacitance. tox = 3.9 x eps0 x area / cox.

4 Change History

Version	Gen. by:	Changes / Remarks
0.1 2025-12-15	**CW**	initial version
0.2 2025-12-15	**CW**	minor updates

5 Known Issues